Introduction

"Figure as far as you can, then add judgment"

The statistical techniques discussed so far are for ***only one variable***. In many research situations one has to consider two variables simultaneously to know whether these ***two variables*** are related linearly. If so, what type of relationship that exists between them. This leads to bivariate (two variables) data analysis namely correlation analysis. If two quantities vary in such a way that movements (upward or downward) in one are accompanied by the movements(upward or downward) in the other, these quantities are said to be co-related or correlated.

The correlation concept will help to answer the following types of questions.

- Whether study time in hours is related with marks scored in the examination?
- Is it worth spending on advertisement for the promotion of sales?
- Whether a woman's age and her systolic blood pressure are related?
- Is age of husband and age of wife related?
- Whether price of a commodity and demand related?
- Is there any relationship between rainfall and production of rice?

4.1 DEFINITION OF CORRELATION

Correlation is a statistical measure which helps in analyzing the interdependence of two or more variables. In this chapter the dependence between only two variables are considered.

1. **A.M. Tuttle** defines correlation as:

   "An analysis of the co-variation of two or more variables is usually called correlation"

2. **Ya-kun-chou** defines correlation as:

   "The attempts to determine the degree of relationship between variables".

Correlation analysis is the process of studying the strength of the relationship between two related variables. High correlation means that variables have a strong linear relationship with each other while a low correlation means that the variables are hardly related. The type and intensity of correlation is measured through the correlation analysis. The measure of correlation is the correlation coefficient or correlation index. It is an absolute measure.

Uses of correlation

- Investigates the type and strength of the relationship that exists between the two variables.
- Progressive development in the methods of science and philosophy has been characterized by the rich knowledge of relationship.

4.2 TYPES OF CORRELATION

1. ***Simple (Linear) correlation*** (2 variables only) : The correlation between the given two variables. It is denoted by r_{xy}
2. ***Partial correlation (more than 2 variables):*** The correlation between any two variables while removing the effect of other variables. It is denoted by $r_{xy.z\ldots}$

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